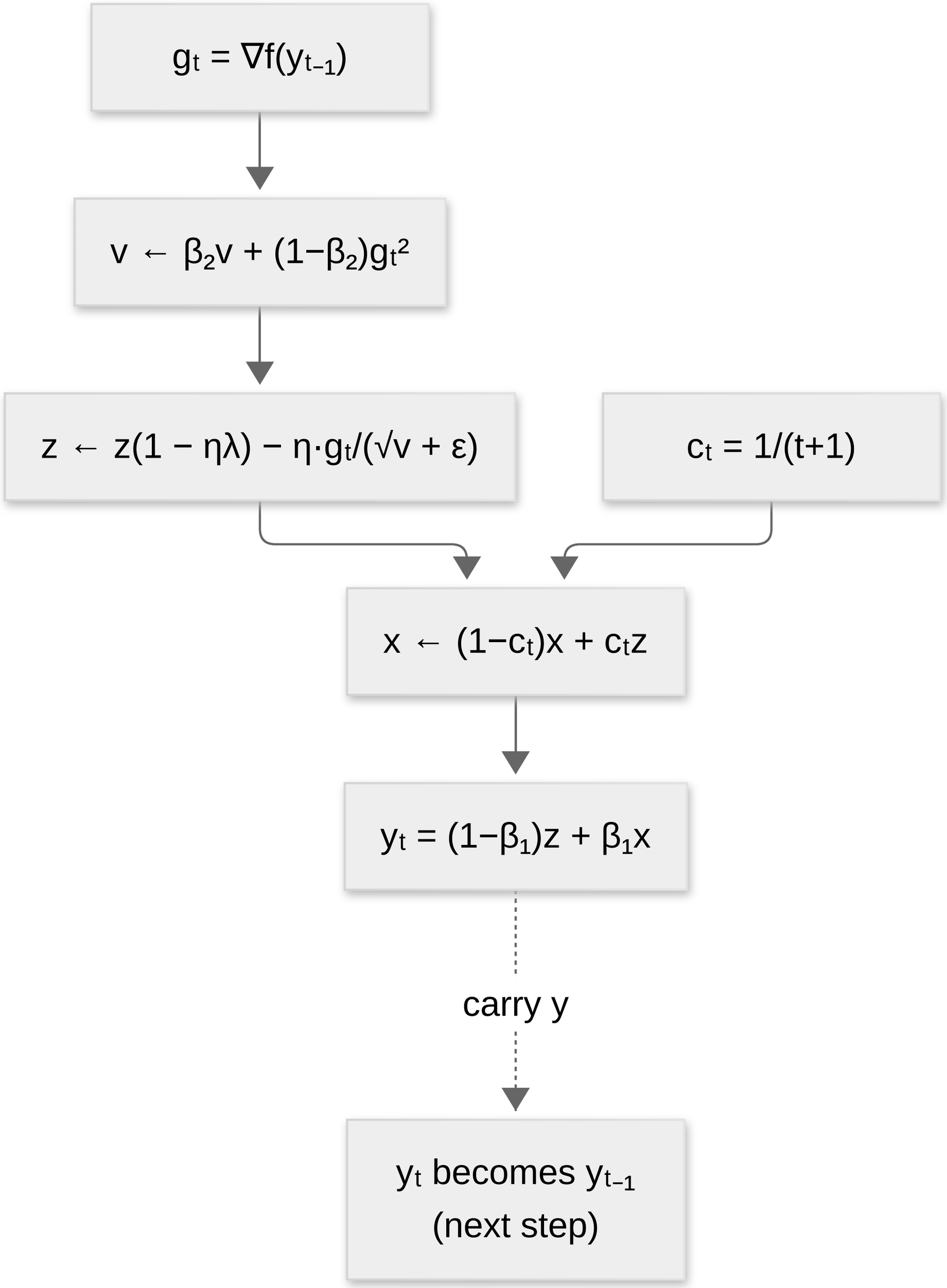

$g_t = \nabla f(y_{t-1})$
$v \leftarrow \beta_2 v + (1-\beta_2) g_t^2$
$z \leftarrow z(1 - \eta\lambda) - \eta \cdot g_t/(\sqrt{v} + \varepsilon)$
$c_t = 1/(t+1)$
$x \leftarrow (1-c_t)x + c_t z$
$y_t = (1-\beta_1)z + \beta_1 x$
carry y
y_t becomes y_{t-1}
(next step)